

1. GENERATE RAW MOCK DATA

```
import pandas as pd
import numpy as np
raw_data = {
    "Employee_Name": [" Alice ", "Bob", "Charlie", "Diana", "Evan"],
    "Department": ["HR", "Engineering", "HR", "Engineering", "Marketing"],
    "Salary": [50000, 85000, np.nan, 92000, 60000],
    "Join_Date": [
        "2022-01-15",
        "2021-06-20",
        "2023-03-11",
        "2020-11-01",
        "2024-02-28",
    ],
}
df = pd.DataFrame(raw_data)
print("--- Raw Data ---")
print(df)
```

```
--- Raw Data ---
  Employee_Name  Department  Salary  Join_Date
0         Alice           HR  50000.0  2022-01-15
1           Bob  Engineering  85000.0  2021-06-20
2        Charlie           HR         NaN  2023-03-11
3         Diana  Engineering  92000.0  2020-11-01
4           Evan  Marketing  60000.0  2024-02-28
```

2. DATA PREPROCESSING

```
df["Employee_Name"] = df["Employee_Name"].str.strip()
median_salary = df["Salary"].median()
df["Salary"] = df["Salary"].fillna(median_salary)
df["Join_Date"] = pd.to_datetime(df["Join_Date"])
df["Years_of_Service"] = 2026 - df["Join_Date"].dt.year
print("---- Preprocessed Data ----")
print(df)
```

```
--- Preprocessed Data ---
  Employee_Name  Department  Salary  Join_Date  Years_of_Service
0         Alice           HR  50000.0  2022-01-15                4
1           Bob  Engineering  85000.0  2021-06-20                5
2        Charlie           HR  72500.0  2023-03-11                3
3         Diana  Engineering  92000.0  2020-11-01                6
4           Evan  Marketing  60000.0  2024-02-28                2
```

3. DATA AGGREGATION

```
dept_summary = (
    df.groupby("Department")
      .agg(
          Avg_Salary=("Salary", "mean"),
          Total_Employees=("Employee_Name", "count")
      )
      .reset_index()
)

print("---- Department Summary ----")
print(dept_summary)
```

```
--- Department Summary ---
  Department  Avg_Salary  Total_Employees
0  Engineering      88500.0                2
1           HR      61250.0                2
2  Marketing      60000.0                1
```

4. DATA VISUALIZATION

```
import matplotlib.pyplot as plt
import seaborn as sns
sns.set_theme(style="whitegrid")
fig, axes = plt.subplots(1, 2, figsize=(14, 5))
sns.barplot(
    data=dept_summary,
    x="Department",
    y="Avg_Salary",
    hue="Department",
    palette="plasma"
)
```

```
plt.xticks(values_u,
            legend=False,
            ax=axes[0])
)
axes[0].set_title("Average Salary by Department")
axes[0].set_xlabel("Department")
axes[0].set_ylabel("Salary ($)")
sns.scatterplot(
    data=df,
    x="Years_of_Service",
    y="Salary",
    hue="Department",
    style="Department",
    s=200,
    ax=axes[1]
)
axes[1].set_title("Salary vs. Years of Service")
axes[1].set_xlabel("Years of Service")
axes[1].set_ylabel("Salary ($)")
plt.tight_layout()
plt.show()
```

